

A32NX flyPad failures

Clearer A32NX failure names and new failures with their FCOM alerts: transponders, RMPs and VHF radios, FWC 1 and 2, engine and APU fires with a real fire detection model and fire detection unit failures, wing tank pumps, the X FEED valve, the centre tank transfer valves, and engine flameouts and seizures with the in-flight relight envelope

Aircraft	A32NX
Where to find it	flyPad > Failures (ATA 21, 22, 23, 26, 28, 31, 34, 72); E/WD, STATUS and SD pages; pedestal ATC and RMP panels; overhead FIRE and FUEL panels; ENG MASTER and ENG MODE controls
Branch	core/failures-a32nx, based on core/gps (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first), with core/fire merged in (the fire failures use its fire warnings, and the engine failures use its ENG FIRE pb LP valve starvation). Only the nine features below belong to this branch; the GPS and fire warning features are described in the core/gps and core/fire manuals.
Status	Transponder, RMP/VHF, FWC, fire and fuel failures tested in the sim by the user: OK (2026-10-04); the fire detection unit failures, the FIRE TEST light logic, the APU fire horn and the centre tank transfer valve failures too (2026-10-04, the horn itself is silent in MSFS 2024); the engine flameout and seizure failures on develop: OK, tests T1-T9 (2026-10-05). Clearer names: not sim-tested (labels only). Nothing is sim-tested on this branch itself.
Pictures	No pictures: the failures page, the ECAM alerts and the panels are the stock FBW displays.

1. Clearer failure names

2. Transponder failures

3. RMP and VHF failures

4. FWC 1 and FWC 2 failures

5. Fire detection model and fire failures
6. Fire detection unit failures, FIRE TEST light logic, APU fire horn

7. Wing tank pump and X FEED valve failures

8. Centre tank transfer valve failures and fuel follow-ups

9. Engine flameout and seizure failures, in-flight relight

1. Clearer failure names

Aircraft:	A32NX		Where:	flyPad > Failures		Status:	Not sim-tested (labels only).
-----------	-------	--	--------	-------------------	--	---------	-------------------------------

What it is for

The failure names were audited ATA by ATA against what the systems code fails. Three names were misleading: the ATA 21 valve is the hot air pressure regulating valve, the FMGC failures only fail the flight guidance part of the FMGC, and the RA failures are antenna failures. Only the labels change.

ATA	Before	Now
21	Trim Air Pressure Regulating Valve	Hot Air Pressure Regulating Valve
22	FMGC 1 / FMGC 2	FMGC 1 (flight guidance) / FMGC 2 (flight guidance)
34	RA 1 Interrupted / RA 2 Interrupted	RA 1 antenna interrupted / RA 2 antenna interrupted
34	RA 1 Direct Coupling / RA 2 Direct Coupling	RA 1 antenna direct coupling / RA 2 antenna direct coupling

How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

How to test

#	Do this	Expected result
---	---------	-----------------

1	Open flyPad > Failures > ATA 21 Air Conditioning.	"Hot Air Pressure Regulating Valve" is listed.
2	Open ATA 22 Auto Flight.	"FMGC 1 (flight guidance)" and "FMGC 2 (flight guidance)".
3	Open ATA 34 Navigation.	"RA 1 antenna interrupted", "RA 2 antenna interrupted", "RA 1 antenna direct coupling", "RA 2 antenna direct coupling".

2. Transponder failures

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 34 Navigation; pedestal ATC/TCAS panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

What it is for

Two new failures, **ATC/XPDR 1** and **ATC/XPDR 2**. MSFS has a single transponder: it stands for the transponder selected on the ATC panel (SYS 1 / SYS 2). When the selected one fails, the sim transponder stops replying and the ATC FAIL light comes on; selecting the other system brings the replies back. With both lost, the TCAS goes to standby. The ECAM alerts and STATUS follow the A320 FCOM PRO-ABN-NAV procedures.

Fix on the way: the ATC/XPDR STBY indication read a second sim transponder that does not exist; the standby state of each transponder now comes from the system selection and the mode selector.

What you should see

Situation	What you should see
ATC/XPDR 1 failed, SYS 1 selected	ATC FAIL light; no transponder reply. E/WD amber NAV ATC/XPDR 1 FAULT with -ATC/XPDR SYS 2 (only while XPDR 1 is selected and XPDR 2 works). STATUS INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.
SYS 2 selected	ATC FAIL off, replies again; the procedure line is done.
ATC/XPDR 2 failed	The same with NAV ATC/XPDR 2 FAULT, -ATC/XPDR SYS 1, INOP SYS ATC/XPDR 2, ADS-B RPTG 2.
Both failed	NAV ATC/XPDR 1+2 FAULT (crew awareness); STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS goes to standby.
A transponder loses its bus (AC ESS SHED for XPDR 1, AC 2 for XPDR 2)	The transponder is lost, but no ATC/XPDR FAULT: a bus loss is an ELEC alert.

Limitations and why

Limitation	Why	What to do
The ATC/XPDR alerts are inhibited during the takeoff and below 800 ft in approach to the landing roll.	The FCOM gives the inhibition as a figure: the one of the RA faults is used (flight phases 3, 4, 5, 7, 8).	Test at the gate or in cruise.

How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate **ATC/XPDR 1** or **ATC/XPDR 2**.
2. Follow the ECAM: select the other system on the ATC/TCAS panel.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, ATC/TCAS panel: SYS 1, mode AUTO (or ON).	No ATC FAIL light.
2	Activate ATC/XPDR 1 .	ATC FAIL light on; master caution and single chime; E/WD NAV ATC/XPDR 1 FAULT with -ATC/XPDR SYS 2. ATC (or a traffic tool) no longer sees your transponder.
3	Open the STATUS page.	INOP SYS: ATC/XPDR 1, ADS-B RPTG 1.
4	Select SYS 2 on the ATC panel.	ATC FAIL off, the transponder replies again, the SYS 2 line is done.
5	Activate ATC/XPDR 2 as well.	NAV ATC/XPDR 1+2 FAULT; STATUS INOP SYS: TCAS, ATC/XPDR 1, ATC/XPDR 2, ADS-B RPTG 1, ADS-B RPTG 2; the TCAS is in standby.
6	Deactivate both failures.	The alerts clear; the TCAS works again.

3. RMP and VHF failures

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 23 Communication; pedestal RMP 1 and RMP 2 | **Status:** Tested in the sim by the user: OK (2026-10-04).

What it is for

Five new ATA 23 failures: **RMP 1**, **RMP 2**, **VHF 1**, **VHF 2** and **VHF 3** (same ids as the A380X ATA 23 failures). A failed RMP goes blank and stops tuning; the other RMP still tunes every VHF (A320 FCOM DSC-23-10-20: "If one RMP fails, the remaining one controls all the VHF transceivers"). A failed VHF switches its MSFS COM radio off: it neither receives nor transmits.

What you should see

Situation	What you should see
RMP 1 (or 2) failed	Both frequency displays blank, the green key lights and SEL off, the keys and knobs do nothing (FCOM DSC-23-60 failure cases).
The other RMP	Still tunes VHF 1, 2 and 3.
VHF 1 (2, 3) failed	COM 1 (2, 3) silent: no reception, no transmission. The RMPs still show and change its frequency.

Limitations and why

Limitation	Why	What to do
No ECAM alert for these failures.	Not part of this change.	Check the RMP and the radio.
A NAV key left on on a failed RMP still holds the FMGC navaid tuning.	As with an RMP switched off: the navaid tuner (shared with the A380X) reads the NAV key only.	Avoid the NAV key before failing an RMP.
No RMP 3 and no audio control panel / AMU failure.	RMP 3 is not in the A32NX cockpit, and the MSFS radios have no audio channel to fail.	None.

How to use

1. Open flyPad > Failures > ATA 23 Communication and activate RMP 1, RMP 2, VHF 1, VHF 2 or VHF 3.
2. With an RMP failed, tune the radios from the other RMP.

How to test

#	Do this	Expected result
1	A32NX powered, an ATIS frequency on VHF 1, audible.	The ATIS is heard.
2	Activate RMP 1 .	RMP 1 goes blank, its key lights go off, its knobs do nothing. The ATIS is still heard (VHF 1 itself works).
3	On RMP 2, press the VHF 1 key and tune another frequency, transfer it.	VHF 1 changes frequency: RMP 2 tunes VHF 1.
4	Deactivate RMP 1. Activate VHF 1 .	COM 1 goes silent; transmitting on VHF 1 is not possible. RMP 1 still shows the frequency.
5	Deactivate VHF 1.	The ATIS is heard again.

4. FWC 1 and FWC 2 failures

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 31 Indicating / Recording System; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

What it is for

Two new failures, **FWC 1** and **FWC 2** (A320 FCOM DSC-31-05-30, PRO-ABN-FWS). With one FWC failed, the other one keeps the warning system working and shows the fault. With both failed, the ECAM alerts, memos, STATUS, MASTER WARN / CAUT lights, aural (including the STALL aural) and the altitude callouts are lost, and the E/WD shows the FWC 1+2 FAULT display. Each FWC also follows its own electrical supply.

What you should see

Situation	What you should see
FWC 1 failed	E/WD FWS FWC 1 FAULT (shown by FWC 2), level 1: no MASTER CAUT, no chime. STATUS: CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.
FWC 2 failed	The same with FWS FWC 2 FAULT and FWC 2.
Both failed	E/WD: FWS FWC 1+2 FAULT , -MONITOR SYS, -MONITOR OVERHEAD PANEL; right side NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. No other alert, memo, STATUS, STS reminder, master light or aural.
One FWC restored	Every active alert is shown again, as a new one.

Limitations and why

Limitation	Why	What to do
FWS FWC 1(2) FAULT is level 1 and inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the level and the inhibition in figures only: those of the other computer faults are used.	Test at the gate or in cruise.

How to use

1. Open flyPad > Failures > ATA 31 Indicating / Recording System and activate FWC 1 and / or FWC 2.
2. With both failed, monitor the systems and the overhead panel yourself.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, no alert. Activate FWC 1 .	E/WD: FWS FWC 1 FAULT, no master caution, no chime.
2	Open the STATUS page.	CAT 3 SINGLE ONLY; INOP SYS: CAT 3 DUAL, FWC 1.
3	Activate FWC 2 as well.	E/WD: FWS FWC 1+2 FAULT, -MONITOR SYS, -MONITOR OVERHEAD PANEL, and NOT AVAIL: ECAM WARN, ALTI ALERT, STATUS, A/CALL OUT, MEMO. The memos are gone.
4	Press the STS key on the ECAM control panel.	No STATUS page (STATUS is in the NOT AVAIL list).
5	Deactivate FWC 2.	FWS FWC 1 FAULT again; the active alerts come back as new ones.

5. Fire detection model and fire failures

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04).

What it is for

Nine new failures: **Engine 1 fire**, **Engine 2 fire**, **APU fire**, and **Engine 1 fire detection loop A**, **Engine 1 fire detection loop B**, **Engine 2 fire detection loop A**, **Engine 2 fire detection loop B**, **APU fire detection loop A**, **APU fire detection loop B**.

Each zone now has the two detection loops and the fire detection unit of the aircraft (A320 FCOM DSC-26-20-10). A fire warning comes when both loops detect the fire, when one loop detects it and the other one is failed, or when both loops break within 5 s (flame effect). The logic moved from the A380X into the common code; the A380X behaviour is unchanged. The fires of the MSFS failures menu still give the warnings, through the loops. The FIRE TEST is unchanged.

- The ENG MASTER panel FIRE light comes on with the engine fire warning (before: FIRE TEST only).
- APU fire on the ground: the APU shuts down and the APU AGENT discharges automatically 3 s after the warning (FCOM DSC-26-20-20).
- Each AGENT discharge has one chance in two to put a flyPad fire out.
- Loop supplies (FCOM PRO-ABN-ELEC INOP SYS): DC BUS 2 for ENG 1 LOOP B and ENG 2 LOOP A, DC ESS BUS for ENG 1 LOOP A and ENG 2 LOOP B, DC BAT BUS for both APU loops. An unpowered loop is a failed loop.

What you should see

Situation	What you should see
Engine 1 fire	ENG 1 FIRE warning (from core/fire), red ENG 1 FIRE pushbutton light and red FIRE light on the ENG MASTER panel.
APU fire on the ground	APU FIRE warning; the APU shuts down; 3 s later the APU AGENT discharges (DISCH light).
AGENT discharged	The fire goes out, or not (one chance in two per discharge).

One loop failed	Amber ENG 1 FIRE LOOP A FAULT (LOOP B, ENG 2, APU FIRE LOOP A / B); STATUS INOP SYS ENG 1 LOOP A (ENG 1 LOOP B, ENG 2 LOOP A / B, APU LOOP A / B). A fire is still detected by the other loop.
Both loops of a zone failed	ENG 1 FIRE DET FAULT (ENG 2, APU FIRE DET FAULT) instead of the two loop cautions; STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).

Limitations and why

Limitation	Why	What to do
FIRE DET FAULT also comes from a failed fire detection unit.	See the next feature, "Fire detection unit failures".	-
The LOOP and DET FAULT cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other crew awareness cautions is used.	Test at the gate or in cruise.
A full MSFS restart is needed after an update.	The systems WASM module and the cockpit behaviour XML change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate a fire or a loop failure.
2. Fight a fire with the ECAM procedure (FIRE pushbutton, AGENT 1, AGENT 2). Deactivating the failure also ends the fire.

How to test

#	Do this	Expected result
1	A32NX on the ground, engines running. Activate Engine 1 fire .	ENG 1 FIRE warning, master warning, continuous chime, red light in the ENG 1 FIRE pushbutton and on the ENG MASTER 1 panel.
2	Follow the ECAM: ENG MASTER 1 OFF, push the ENG 1 FIRE pushbutton, press AGENT 1.	AGENT 1 DISCH; the fire goes out or not (one chance in two). If not, AGENT 2 gives another chance.
3	Deactivate the failure, reset the FIRE pushbutton. With the APU running, activate APU fire .	APU FIRE warning; the APU shuts down; about 3 s later the APU AGENT discharges by itself.
4	Deactivate it. Activate Engine 2 fire detection loop A .	Amber ENG 2 FIRE LOOP A FAULT; STATUS INOP SYS ENG 2 LOOP A.
5	Activate Engine 2 fire with loop A still failed.	ENG 2 FIRE warning: loop B alone detects the fire.
6	Deactivate the fire, activate Engine 2 fire detection loop B as well.	The loop cautions are replaced by ENG 2 FIRE DET FAULT; STATUS INOP SYS FIRE DET 2.
7	Optional: start an engine fire from the MSFS failures menu.	The ENG FIRE warning still comes (through the loops).

6. Fire detection unit failures, FIRE TEST light logic, APU fire horn

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 26 Fire Protection; overhead FIRE panel; ENG MASTER panel; E/WD, STATUS | **Status:** Tested in the sim by the user: OK (2026-10-04); the horn sound itself is silent in MSFS 2024. Rust tests (fire_protection), unit tests: FireDetectionFaults.spec.ts, EngineMasterFireLight.spec.ts.

What it is for

Three follow-ups of the fire detection model above:

- **Fire detection unit failures:** three new failures, **Engine 1 fire detection unit (FDU)**, **Engine 2 fire detection unit (FDU)** and **APU fire detection unit (FDU)**. A failed FDU gives the FIRE DET FAULT caution of its zone even with healthy loops, and detects no fire. Clearing the failure with a fire present gives the fire warning at once.
- **ENG MASTER panel FIRE light during the FIRE TEST:** the A320 FCOM (DSC-26-20-20, ENG 1(2) FIRE TEST pb) says that the FIRE lights do not come on with an engine 1 loop A fault or an engine 2 loop B fault. The light now stays dark in those cases (a loop fault is its failure or the loss of its DC ESS supply).
- **APU fire external horn:** an APU fire on the ground sounds the horn in the nose gear bay (FCOM DSC-26-20-20 APU FIRE light, DSC-49-10-20), on the same sound as the MECH call; not during the FIRE TEST.

What you should see

Situation	What you should see
Engine 1 FDU failed	Amber ENG 1 FIRE DET FAULT (ENG 2 FIRE DET FAULT, APU FIRE DET FAULT for the other units), no LOOP FAULT caution (a failed FDU cannot tell which loop is failed); STATUS INOP SYS FIRE DET 1 (FIRE DET 2, APU FIRE DET).
A fire in a zone with its FDU failed	No fire warning.
FDU failure cleared while the fire burns	The fire warning comes at once.
ENG 1 FIRE TEST with Engine 1 fire detection loop A failed	The ENG MASTER 1 FIRE light stays dark; the rest of the test as before.
ENG 2 FIRE TEST with Engine 2 fire detection loop B failed	The ENG MASTER 2 FIRE light stays dark.
APU fire on the ground	The external horn variable is set (L:A32NX_FIRE_APU_EXTERNAL_HORN); the horn sound plays at the nose gear.

Limitations and why

Limitation	Why	What to do
The APU fire horn is not heard in MSFS 2024.	The Asobo horn sound it uses is silent in MSFS 2024 (the MECH call too).	Check the L:var with a SimConnect / variable tool.
No APU SHUT OFF pb and no APU FIRE light on the external power panel.	The exterior model has no such panel.	None.
A full MSFS restart is needed after an update.	The systems WASM module, the cockpit behaviour XML and the sound.xml change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 26 Fire Protection and activate an "... fire detection unit (FDU)" failure.
2. Follow the ECAM (crew awareness); deactivate the failure to restore the unit.

How to test

#	Do this	Expected result
1	A32NX at the gate, powered, no failure. Activate Engine 1 fire detection unit (FDU) .	Amber ENG 1 FIRE DET FAULT, no ENG 1 FIRE LOOP A / B FAULT; STATUS INOP SYS FIRE DET 1.
2	Also activate Engine 1 fire .	No ENG 1 FIRE warning.

3	Deactivate Engine 1 fire detection unit (FDU) .	The ENG 1 FIRE warning comes at once. Deactivate the fire.
4	Repeat with Engine 2 fire detection unit (FDU) and APU fire detection unit (FDU) .	ENG 2 FIRE DET FAULT / STATUS FIRE DET 2; APU FIRE DET FAULT / STATUS APU FIRE DET.
5	Activate Engine 1 fire detection loop A , then press and hold the ENG 1 FIRE TEST pb.	The ENG 1 FIRE pushbutton lights as in the normal test, but the FIRE light on the ENG MASTER 1 panel stays dark.
6	Deactivate it, activate Engine 2 fire detection loop B , hold the ENG 2 FIRE TEST pb.	The ENG MASTER 2 FIRE light stays dark. With loop A of engine 2 failed instead, it comes on.
7	On the ground with the APU running, activate APU fire .	APU FIRE warning; L:A32NX_FIRE_APU_EXTERNAL_HORN is 1 (the horn itself is silent in MSFS 2024).

7. Wing tank pump and X FEED valve failures

Aircraft: A32NX Where: flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL Status: Tested in the sim by the user: OK (2026-10-04), after the fix that holds the MSFS pump / valve switch.
--

What it is for

Five new failures: **L TK pump 1**, **L TK pump 2**, **R TK pump 1**, **R TK pump 2** and **X FEED valve jammed in place**. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, the jammed valve is held where it was. The FAULT lights, the SD FUEL page and the cautions compare the selection with the real state (A320 FCOM PRO-ABN-FUEL).

- SD FUEL page: amber LO only for a pump that is ON with a low pressure; the X FEED valve is amber when it disagrees with its pushbutton (FCOM colours).
- The FUEL X FEED memo now needs the X FEED pushbutton ON and the valve not fully closed.

What you should see

Situation	What you should see
L TK pump 1 failed, pushbutton ON	FAULT light; amber LO on the SD FUEL page (page called). E/WD FUEL L TK PUMP 1 LO PR with -L TK PUMP 1 OFF (shown while the pushbutton is ON). STATUS INOP SYS L TK PUMP 1.
Both pumps of one wing failed	FUEL L TK PUMP 1+2 LO PR (R for the right wing): .IF NO FUEL LEAK: -FUEL X FEED ON (above FL 150 only), -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, .WHEN L TK FUEL RQRD: L TK FEED GRVTY ONLY, -FUEL X FEED OFF (below FL 150, X FEED ON), PROC:GRVTY FUEL FEEDING, AVOID NEGATIVE G FACTOR. Each action line goes once done. STATUS INOP SYS L TK PUMPS.
X FEED valve jammed, pushbutton moved	The valve stays: amber on the SD FUEL page, FUEL X FEED VALVE FAULT (crew awareness), STATUS INOP SYS FUEL X FEED. With the valve stuck closed, no FUEL X FEED memo.

Limitations and why

Limitation	Why	What to do
------------	-----	------------

The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other single system faults is used.	Test at the gate or in cruise.
The centre tank has no pump failure.	The A320neo centre tank has no pumps: its fuel is moved by the jet pumps through the CTR TK L / R XFR valves, whose jams are a feature of their own (below).	None.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or the X FEED valve failure.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.

How to test

#	Do this	Expected result
1	A32NX at the gate, APU or engines running, all fuel pumps ON. Activate L TK pump 1 .	After a few seconds: FAULT light in the L TK PUMP 1 pushbutton, SD FUEL page with amber LO, master caution, E/WD FUEL L TK PUMP 1 LO PR with -L TK PUMP 1 OFF.
2	Set the L TK PUMP 1 pushbutton OFF. Open STATUS.	The FAULT light and the OFF line go; no LO on the SD; STATUS INOP SYS L TK PUMP 1.
3	Set it ON again and also activate L TK pump 2 .	FUEL L TK PUMP 1+2 LO PR with its procedure (on the ground, below FL 150: .IF NO FUEL LEAK; -ENG MODE SEL IGN, -L TK PUMP 1 OFF, -L TK PUMP 2 OFF, ...); STATUS INOP SYS L TK PUMPS.
4	Deactivate both. With the X FEED valve closed, activate X FEED valve jammed in place , then set the X FEED pushbutton ON.	The valve stays closed: amber X FEED valve on the SD, FUEL X FEED VALVE FAULT, STATUS INOP SYS FUEL X FEED, no FUEL X FEED memo.
5	Deactivate the failure.	The valve opens; the FUEL X FEED memo comes; the caution clears.

8. Centre tank transfer valve failures and fuel follow-ups

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL; flyPad checklists | **Status:** Tested in the sim by the user: OK (2026-10-04). Unit tests: FuelPumpValveFaults.spec.ts, FuelPumpsAndValves.spec.ts, FuelPumpValveIndications.spec.ts, CheckItemStates.spec.ts.

What it is for

Two new failures, **CTR TK L XFR valve jammed in place** and **CTR TK R XFR valve jammed in place**: the centre tank transfer valve of that side stays open or closed, whatever its CTR TK XFR pushbutton commands. The FWC raises the A320 FCOM PRO-ABN-FUEL cautions: FUEL CTR L(R) XFR FAULT (valve not fully closed, or not fully open) and, with both valves stuck the same way, FUEL CTR L + R XFR FAULT, which replaces the two single ones. The SD FUEL page shows the valve amber and boxes the centre tank quantity.

- The centre tank fuel only transfers while an inner tank pump of the same wing runs: the jet pumps are driven by the wing tank pumps (FCOM DSC-28-10). Switching the wing pumps off stops the centre tank transfer.
- flyPad checklist: the securing / parking checklist item **FUEL PUMPS ... OFF** now reads the four wing tank pump pushbuttons, not whether the pumps run (the CTR TK XFR pushbuttons are items of their own in the FCOM).

What you should see

Situation	What you should see
L valve stuck open (CTR TK L XFR pb OFF)	E/WD FUEL CTR L XFR FAULT : -CTR TK L XFR ... OFF, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF, .WHEN CTR TK EMPTY: -L TK PUMP 1 ... ON, -L TK PUMP 2 ... ON, -FUEL X FEED ... OFF (each action line goes once done). STATUS INOP SYS CTR TK L XFR.
L valve stuck closed (CTR TK L XFR pb ON)	FUEL CTR L XFR FAULT : -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: -FUEL X FEED ... ON, -L TK PUMP 1 / 2 ... OFF, .WHEN CTR TK EMPTY: pumps ON, X FEED OFF. STATUS INOP SYS CTR TK L XFR.
Both valves stuck open	FUEL CTR L+R XFR FAULT : -CTR TK L XFR ... OFF, -CTR TK R XFR ... OFF; STATUS INOP SYS CTR TK XFR.
Both valves stuck closed	FUEL CTR L+R XFR FAULT : -FUEL MODE SEL ... MAN, .IF UNSUCCESSFUL: CTR AVAIL BY GRAVITY, 2T (4400LBS) UNUSABLE; STATUS: CTR TK USABLE BY GRAVITY, 2T (4400LBS) UNUSABLE, INOP SYS CTR TK XFR.
SD FUEL page	The jammed valve amber; the centre tank quantity boxed.

Limitations and why

Limitation	Why	What to do
The cautions are inhibited during the takeoff and the landing (phases 3, 4, 5, 7, 8).	The FCOM gives the inhibition as a figure: that of the other fuel cautions is used.	Test at the gate or in cruise.
No gravity transfer from the centre tank.	Not simulated: with both valves stuck closed, the centre tank fuel stays there.	Plan with the centre tank fuel as unusable.

How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate **CTR TK L XFR valve jammed in place** or **CTR TK R XFR valve jammed in place**.
2. Move the CTR TK XFR pushbutton (or let the automatic control move the valve), then follow the ECAM with the overhead FUEL panel.

How to test

#	Do this	Expected result
1	A32NX at the gate with fuel in the centre tank, engines or APU running, all FUEL pushbuttons ON. Activate CTR TK L XFR valve jammed in place while the valve is open, then set the CTR TK L XFR pushbutton OFF.	The valve stays open: master caution, E/WD FUEL CTR L XFR FAULT with -CTR TK L XFR ... OFF done, -FUEL X FEED ... ON, -L TK PUMP 1 ... OFF, -L TK PUMP 2 ... OFF; SD FUEL with the valve amber.
2	Open the STATUS page.	INOP SYS: CTR TK L XFR.
3	Also activate CTR TK R XFR valve jammed in place and set the CTR TK R XFR pushbutton OFF.	The two single cautions are replaced by FUEL CTR L+R XFR FAULT; STATUS INOP SYS CTR TK XFR.
4	Deactivate both failures, CTR TK XFR pushbuttons ON. Set L TK PUMP 1 and 2 OFF.	The left centre tank transfer stops (no jet pump drive) while the right one goes on.
5	Set every wing tank pump pushbutton OFF and open the flyPad checklist with the FUEL PUMPS ... OFF item.	The item is checked from the four wing tank pump pushbuttons.

9. Engine flameout and seizure failures, in-flight relight

Aircraft: A32NX | **Where:** flyPad > Failures > ATA 72 Engine; ENG MASTER 1 / 2, ENG MODE selector; E/WD, STATUS, SD ENG | **Status:** Tested in the sim by the user on develop: OK, tests T1-T9 (2026-10-05). Rust tests (engine_failure.rs of systems and a320_systems, full-aircraft tests incl. the ENG FIRE pb LP valve starvation), unit tests: EngineFailAlerts.spec.ts, A32NX_FADEC.spec.ts, Eng.spec.ts, a320.spec.ts. Not sim-tested on this branch.

What it is for

Four new failures in a new flyPad chapter, ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew can relight the engine with the ENG MASTER OFF then ON, inside the in-flight relight envelope. A **seizure** also stops the core: no relight while the failure is active. The FWC gives the A320 FCOM ENG 1(2) FAIL and ENG 1(2) SHUT DOWN alerts with their STATUS, and the flyPad failures page now describes the ATA 28 Fuel and ATA 72 Engine chapters like the others.

Id	flyPad failure (ATA 72 Engine)	What the code does
72000	Engine 1 flameout (crew relight possible)	Cuts the engine 1 fuel (latched). The engine runs down and windmills; the FADEC does not relight it. ENG MASTER 1 OFF then ON inside the relight envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout.
72001	Engine 2 flameout (crew relight possible)	The same for engine 2.
72010	Engine 1 seizure (no relight)	Cuts the fuel; the core (N2) stops in about 3 s while N1 windmills. No relight while the failure is active; once it is cleared the engine relights by the normal procedure.
72011	Engine 2 seizure (no relight)	The same for engine 2.

The **relight envelope** comes from the A320neo QRH chart ENG RELIGHT IN FLIGHT (IndiGo FCOM, PDF page 4761, PW1100G engines; FBW models the LEAP-1A, for which no chart is available). Vc is the indicated airspeed, Zp the pressure altitude.

Zone	Where	Needs
3, stabilized windmill relight	260-350 kt up to 20 000 ft; 281-350 kt from 20 000 to 26 000 ft (the ceiling falls to 24 000 ft at 350 kt)	ENG MODE IGN, ENG MASTER OFF then ON
1, starter assisted relight	150-260 kt up to 15 000 ft (the lowest speed rises to 180 kt at 20 000 ft); 200-280 kt from 20 000 to 28 000 ft	ENG MODE IGN, ENG MASTER OFF then ON, and starter air (APU BLEED, or X BLEED OPEN with the other engine)
2, windmill quick relight	230-260 kt up to 13 000 ft	ENG MASTER OFF then ON within 20 s, with N2 above 15 %, at any ENG MODE position

- Above these limits (e.g. in cruise at FL 350) no relight lights up.
- A windmilling engine (no combustion, in flight) keeps $N2 = 8.5 \% \times IAS / 260 \text{ kt}$ and $N1 = 1.5 \times N2$ (at 300 kt: N2 9.8 %, N1 14.7 %); on the ground it runs down to 0. Its fuel flow shows 0 and its oil pressure falls with N2.
- During an in-flight relight both igniters are supplied (A320 FCOM DSC-70-80-30), whatever the ENG MODE selector: the SD ENG page shows A B.
- The SD ENG page reads the N2 the FADEC shows, so a seized core reads 0 there as on the E/WD.
- ENG ALL ENGINES FAILURE gets its FCOM flight phase inhibition (phases 1-4 and 8-10).

What you should see

Situation	What you should see
-----------	---------------------

Flameout in flight	N2, EGT and FF run down, FF 0, N1 / N2 settle at the windmill values. Amber ENG 1 FAIL , single chime, MASTER CAUT, SD ENG page: -ENG MODE SEL IGN, -THR LEVER 1 IDLE, .IF NO RELIGHT AFT 30S;; 30 s after the alert -ENG MASTER 1 OFF, .IF DAMAGE: -ENG 1 FIRE P/B PUSH, -AGENT1 AFTER 10S DISCH, .IF NO DAMAGE: -ENG 1 RELIGHT CONSIDER (A320 FCOM PRO-ABN-ENG ENG 1(2) FAIL). LAND ASAP amber. The ENG 1 FAULT light stays off (FCOM: a flameout does not light it).
ENG MASTER 1 OFF in flight (after a failure or not)	ENG 1 FAIL goes, ENG 1 SHUT DOWN comes: .IF NO FUEL LEAK;; - IMBALANCE MONITOR, -TCAS MODE SEL TA (with the wing anti-ice on: the pack of the affected side OFF and X BLEED OPEN; with the ENG FIRE pb pushed: X BLEED SHUT, WING ANTI ICE OFF, AVOID ICING CONDITIONS). STATUS: IF PERF PERMITS: X BLEED OPEN, IF NO ENG 1 DAMAGE: CONSIDER ENG 1 RELIGHT, CAT 3 SINGLE ONLY, ONE PACK ONLY IF WAI ON; INOP SYS: CAT 3 DUAL, ENG 1 BLEED, PACK 1, MAIN GALLEY, GEN 1, G ENG 1 PUMP (WING A. ICE with the ENG FIRE pb pushed).
Relight inside the envelope	Light-up: FF, N2 and EGT rise, A B igniters on the SD ENG page, AVAIL; the engine alerts and LAND ASAP clear.
Relight outside the envelope	No light-up: N2 stays at the windmill value, FF 0.
Seizure	N2 to 0 in about 3 s, N1 windmills, oil pressure 0 (red), ENG 2 FAIL as above; GEN 2 and the yellow engine pump lost.
Both engines failed	Red ENG ALL ENGINES FAILURE; the single ENG 1 / ENG 2 FAIL alerts do not show while it is active. Relighting one engine brings the single alert of the other one.
On the ground	ENG 1 FAIL with the ground lines (-THR LEVER 1 IDLE, -ENG MASTER 1 OFF, .IF DAMAGE: ..., -AGENT 1 DISCH, .IF NO DAMAGE: ...), no LAND ASAP; N1 / N2 run down to 0. ENG MASTER OFF gives no ENG SHUT DOWN in flight phases 1-2 and 9-10 (FCOM: phases 3-8 only).

Limitations and design choices

Limitation	Why	What to do
The relight envelope applies to every in-flight relight, also after a crew shutdown without any failure.	Design choice: MSFS would otherwise relight anywhere; the QRH chart is the limit of any relight.	Relight inside the envelope (descend and adjust the speed).
The envelope is the A320neo PW1100G chart.	FBW models the LEAP-1A; no LEAP-1A relight chart is available.	-
The windmill N1 / N2 values are design values.	The chart only marks zone 3 "N2 > 8.5 %"; N2 is taken proportional to the airspeed, 8.5 % at 260 kt, and N1 = 1.5 x N2.	-
The seizure stops the core in about 3 s.	Design choice: the FCOM gives no rate.	-
ENG 1(2) FAIL does not come back after the ENG MASTER OFF, even if the relight does not light up.	Design choice: the MASTER OFF disarms the alert; ENG SHUT DOWN is shown meanwhile.	After a failed attempt (no light-up within 30 s), MASTER OFF and try again.

While ENG ALL ENGINES FAILURE is active, the single engine alerts are not shown.	Design choice: its procedure replaces them.	-
The HP fuel valve opening at N2 > 15 % is not modelled.	The fuel comes back when the engine lights up inside the envelope.	-
On this branch the oil pressure of a running engine is the stock FBW one.	The FADEC oil model of develop is a fork-only change (47e00cee8) that is not on this base; only the shutdown oil pressure (following N2) comes with this feature.	-
A full MSFS restart is needed after an update.	The systems WASM module and the FADEC WASM module change.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM (ENG FAIL, then ENG SHUT DOWN). To relight, get inside the envelope (table above), ENG MODE IGN, ENG MASTER OFF for 30 s, then ON. For a quick relight in zone 2, set the MASTER OFF then ON at once (within 20 s, N2 above 15 %).
3. Clear the failure in the flyPad after the test; a seizure must be cleared before a relight.

How to test

From the test plan `reports/test-plan-engine-failures-a1.md`. Setups: S1 cruise 60 t, FL 350, M.78; S2 60 t, 10 000 ft, 300 kt; S3 60 t, 15 000 ft, 210 kt, APU running with APU BLEED ON; S4 on the ground, both engines at idle.

#	Do this	Expected result
1	T1 (S1): activate Engine 1 flameout (crew relight possible) .	ENG 1 FAIL with its lines, LAND ASAP, windmill N1 / N2, FF 0, oil pressure falling with N2; the 30 s lines come after 30 s. ENG MODE IGN and THR LEVER 1 IDLE clear their lines.
2	T1: after 30 s, ENG MASTER 1 OFF; then MASTER ON at FL 350.	ENG 1 SHUT DOWN with its STATUS and INOP SYS; at FL 350 no light-up (outside the envelope), A B igniters on the SD ENG page.
3	T2: descend to 20 000 ft or below at 280-300 kt; ENG MODE IGN, MASTER 1 OFF 30 s, ON.	Light-up at once; the engine back at idle in about 30-60 s; all engine 1 alerts and LAND ASAP clear.
4	T3 (S3): flameout; MASTER 1 OFF; APU BLEED OFF, X BLEED not open; 30 s; MASTER 1 ON.	No light-up (210 kt is below zone 3, no starter air). MASTER OFF, APU BLEED ON, 30 s, MASTER ON: the start valve opens and the engine lights up.
5	T4 (S2 slowed to 245 kt below 13 000 ft, ENG MODE NORM): flameout, then at once MASTER 1 OFF then ON within 20 s.	Quick relight at NORM; the ENG MODE knob stays at NORM and still works afterwards. Waiting until N2 is below 15 % before the MASTER OFF / ON: no light-up.
6	T5 (S2): activate Engine 2 seizure (no relight) .	N2 to 0 in about 3 s, N1 windmills, oil pressure 0, ENG 2 FAIL; MASTER 2 OFF 30 s / ON at 300 kt: no light-up. Clear the seizure, MASTER 2 OFF 30 s / ON: light-up.
7	T6 (S2, optional): activate both flameouts.	ENG ALL ENGINES FAILURE without the single ENG FAIL alerts; relighting engine 1 brings ENG 2 FAIL.
8	T7 (S1): no failure, MASTER 1 OFF at FL 350, then ON with ENG MODE IGN.	ENG 1 SHUT DOWN; no light-up at FL 350; at FL 200 / 290 kt, MASTER OFF 30 s then ON: light-up.

9	T8 (S4): activate Engine 1 flameout on the ground.	ENG 1 FAIL with the ground lines, no LAND ASAP, N1 / N2 to 0; MASTER 1 OFF gives no ENG 1 SHUT DOWN; a normal start works.
10	T9: normal flight (starts, single-engine taxi, takeoff to descent, shutdown); on the ground at idle, push the ENG 1 FIRE pb with MASTER 1 ON.	No ENG FAIL / SHUT DOWN alert in the normal flight. With the FIRE pb: ENG 1 SHUT DOWN at once, no ENG 1 FAIL; the engine stops after about 100 s on its residual fuel (LP valve starvation of core/fire).